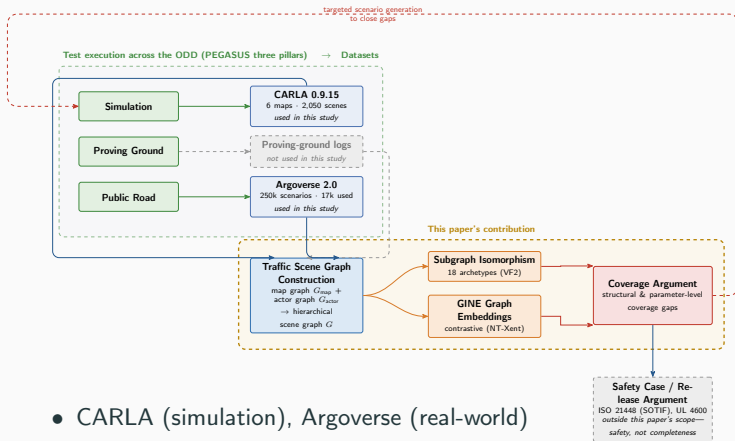


# A Graph-based Framework for Coverage Analysis in Autonomous Driving

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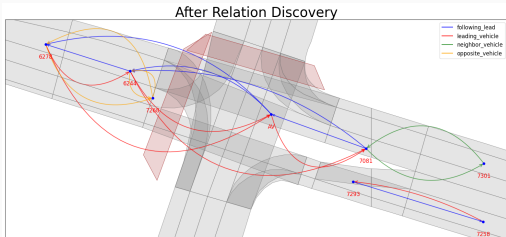
Thomas Mühlenstädt   Marius Bause

# Verification and Validation Workflow

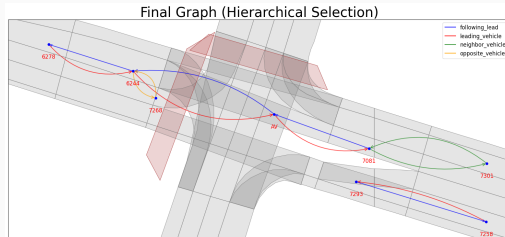


- CARLA (simulation), Argoverse (real-world)
- Ours: coverage argument via graph representation

# Hierarchical Scene Graph Construction



After relation discovery

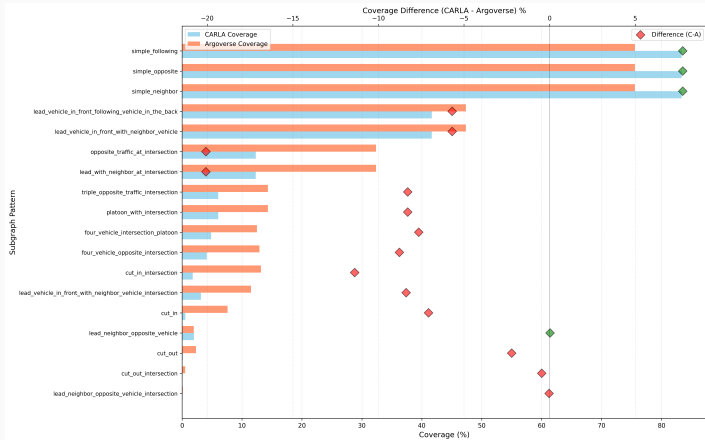


After hierarchical selection

- Map graph edges types included: following, neighbor, opposite
- Lane-change indicator: lane change within  $\Delta t = 1s$
- Phase 1 – Relation Discovery → Phase 2 – Hierarchical Construction by selecting by edge type and distance

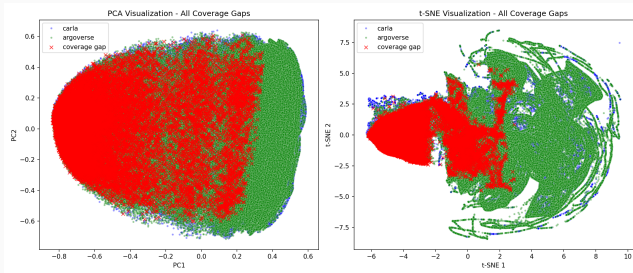
# Top-Down Analysis: Scenario Archetype Coverage Gaps

- Archetypes: 18 types user-defined (can be detected by algorithm)
- Subgraph matching via the **VF2** algorithm — near-linear in the number of nodes of the scene graph (sparse graphs)
- Result: missing archetypes reveal coverage gaps



# Bottom-Up Analysis: Embedding Space Coverage Gaps

- **GINE**: edge-aware graph embeddings
- Self-supervised contrastive training (CARLA + Argoverse)
- Plausibility check: nearest neighbours match
- First 2 principal components: 40% variance
- 23,148 Argoverse graphs flagged as gaps

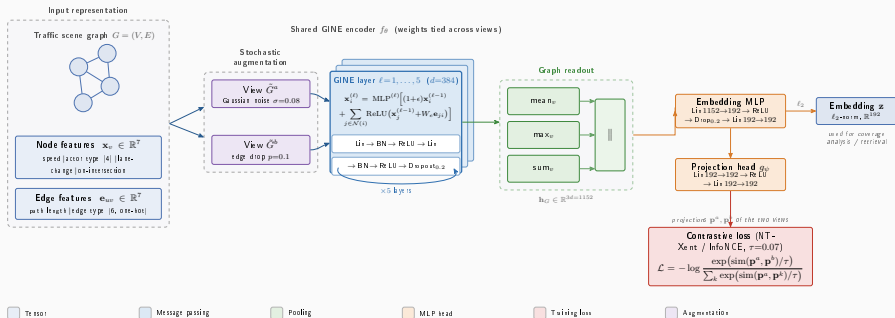


# Summary and Future Work

- Graph-based framework: hierarchical scene graphs combining map topology with actor relationships
- Two complementary coverage methods:
  - subgraph isomorphism against archetypes → interpretable, pattern-based coverage
  - GINE embeddings via contrastive learning → similarity-based coverage, clustering, anomaly detection
- Validated on Argoverse 2.0 and CARLA: reveals structural, parametric, and co-occurrence coverage gaps in simulation
- Scales across scenarios without scenario-specific handling
- Naturally accommodates varying numbers of actors
- **Future work:**
  - multi-timestep temporal graphs
  - automatically extracting new archetype candidates from dense, unmatched embedding regions
- Code: [https://github.com/tmuehlen80/graph\\_coverage](https://github.com/tmuehlen80/graph_coverage)

Backup slides with additional graphics

# Supplement: GINE Encoder Architecture

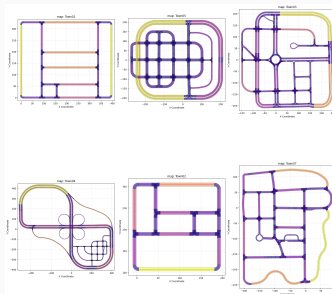


# Supplement: Scenario Archetype Overview

| Group             | Archetype                            | Actors | Edges | Edge Types     |
|-------------------|--------------------------------------|--------|-------|----------------|
| Simple (2-actor)  | Simple Following                     | 2      | 2     | fl, lv         |
|                   | Simple Opposite                      | 2      | 2     | ov             |
|                   | Simple Neighbor                      | 2      | 2     | nv             |
| Complex (3-actor) | Lead + Neighbor (intersection)       | 3      | 4     | fl, lv, nv     |
|                   | Lead + Neighbor                      | 3      | 4     | fl, lv, nv     |
|                   | Cut-in                               | 3      | 4     | fl, lv         |
|                   | Cut-in (intersection)                | 3      | 4     | fl, lv         |
|                   | Platoon (intersection)               | 3      | 4     | fl, lv         |
|                   | Opposite Traffic (intersection)      | 3      | 4     | fl, lv, ov     |
|                   | Lead + Neighbor at Intersection      | 3      | 4     | fl, lv, nv     |
|                   | Triple Opposite (intersection)       | 3      | 4     | ov             |
|                   | Lead + Following in Back             | 3      | 4     | fl, lv         |
| Complex (4-actor) | Cut-out                              | 4      | 6     | fl, lv, nv     |
|                   | Cut-out (intersection)               | 4      | 6     | fl, lv, nv     |
|                   | 4-Vehicle Platoon (intersection)     | 4      | 6     | fl, lv         |
|                   | 4-Vehicle Opposite (intersection)    | 4      | 6     | fl, lv, ov     |
| Complex (5-actor) | Lead + Neighbor + Opposite           | 5      | 8     | fl, lv, nv, ov |
|                   | Lead + Neighbor + Opposite (inters.) | 5      | 8     | fl, lv, nv, ov |

Edge types: *fl* = following\_lead, *lv* = leading\_vehicle, *nv* = neighbor\_vehicle, *ov* = opposite\_vehicle.

# Supplement: CARLA and Argoverse 2.0 Datasets



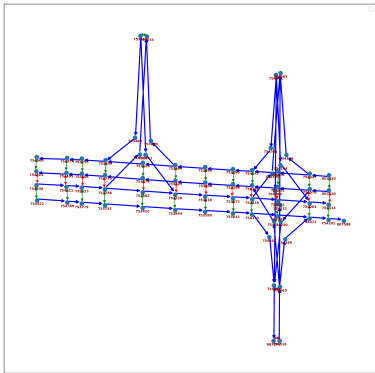
## Argoverse 2.0

- Real-world data, 6 US cities: Austin, Detroit, Miami, Palo Alto, Pittsburgh, Washington D.C.
- LiDAR, ring camera imagery, HD vector maps
- Motion Forecasting Dataset: 250,000 scenarios, 11 s each
- Subset used here: **17,000 scenes** (train partition)

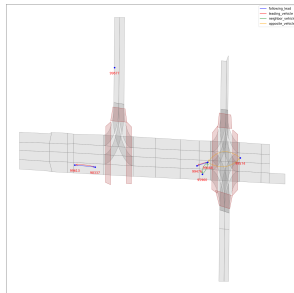
## CARLA

- Version 0.9.15; maps Town01, Town02, Town03, Town04, Town05, Town07
- Custom generation script: trucks, motorcycles, cars with varying speed/following/lane-change behavior
- 20–60 vehicles per simulation run
- **2,050 scenes**, 11 s each

# Supplement: Map and Traffic Scene Graphs



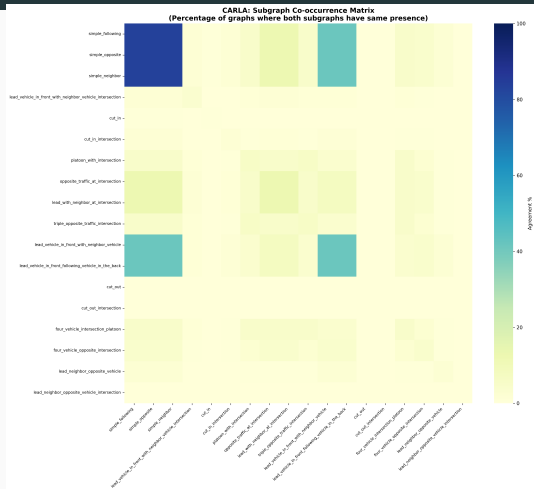
Map graph



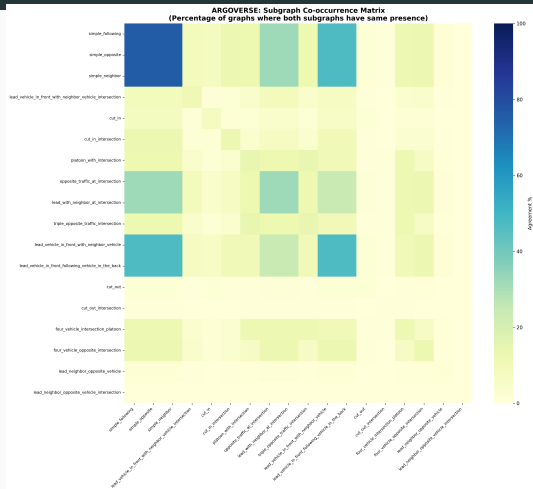
Traffic scene graph at one timestep

The map graph (left) encodes spatial relationships between lanes; the traffic scene graph (right) shows actors and their relationships. Actors can be disconnected if they exceed the distance thresholds.

# Supplement: Archetype Co-occurrence



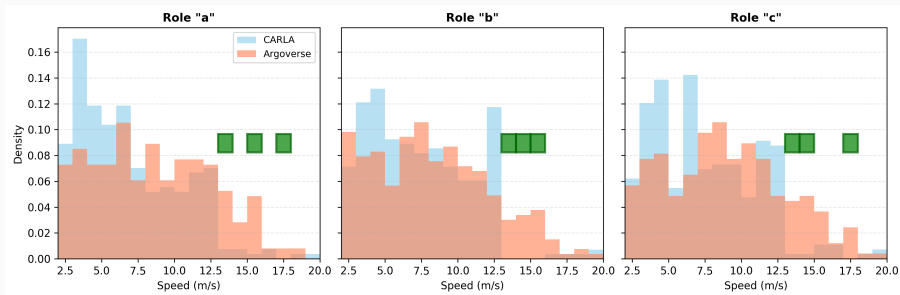
CARLA



Argoverse

Percentage of scenes in which pairs of archetypes co-occur, for CARLA and Argoverse.

## Supplement: Definition Holes via Parameter Distribution



Green

rectangles mark speed ranges where Argoverse shows sufficient density but CARLA is nearly empty, for the `lead_vehicle_in_front_with_neighbor_vehicle` scenario.